alternative is to create indexes *explicitly;* we have much more control over customizing indexes when we create them via that route. Let's look at both approaches:

- *Implicit (automatic) creation*—When indexing a document for the first time, if the index doesn't exist, Elasticsearch creates it implicitly with default settings. This method of index creation usually works well, but care should be taken when using this approach in production because incorrect or unoptimized indexes will bring unexpected consequences to the running system.

  Elasticsearch uses dynamic mapping to deduce field types when creating the mapping schema with this method. Unfortunately, the mapping definitions produced are not foolproof; for example, data in a non-ISO date format (`dd-MM-yyyy` or `MM-dd-yyyy`) is determined to be a `text` field rather than a `date` data type.
- *Explicit (manual) creation*—Choosing this approach lets us control index creation so we can customize the index as required. We can configure the index with a mapping schema doctored by resident data architects, allocate shards based on current and projected storage expectations, and so on.

  Elasticsearch provides a set of index creation APIs that help us create indexes with personalized configurations. We can take advantage of these APIs when creating indexes up front so the indexes are optimized for storage and data retrieval. The APIs provide great flexibility; for example, we can create an individual index with features such as appropriately sized shards, applicable mapping definitions, multiple aliases, and more.

NOTE To control automatic index creation, we can turn off creating indexes by setting the `action.auto_create_index` flag to `false` via the cluster settings API or setting this property in config/elasticsearch.yml. By default, this flag is set to `true`. We use this feature shortly.

6.2.1 Creating indexes implicitly (automatic creation)

When we index a document for the first time, Elasticsearch doesn't complain about a nonexistent index; instead, it happily creates one for us. When an index is created this way, Elasticsearch uses default settings such as setting the number of primary and replica shards to one. To demonstrate, let's quickly index a document with `car` information using the document API. (We created the `cars` index in chapter 4. We delete and re-create it to start fresh in this chapter.)

Listing 6.1 First document containing car data

```
DELETE cars

PUT cars/_doc/1
{
  "make":"Maserati",
  "model":"GranTurismo Sport",
```

Deletes the cars index so we start from scratch

The index doesn't exist but is created the first time a document is indexed. By indexing a document, we create the cars index implicitly.